Treatment of Gastrointestinal infections

3rd Stage Students

Hawler Medical University

College of Medicine

Department of Pharmacology

Dr. Jangi Salai

MD, MSc, PhD

Onsultant Oncologist and Internist

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Objective

- Causes of Gastrointestinal infections.
- The symptoms of Gastrointestinal infections.
- Treatment of viral Gastrointestinal infection.
- Treatment of bacterial Gastrointestinal infection.
- Antibiotic's:

✓ Pharmacokinetics,

✓ Adverse effects and

✓ contraindications

- **Gastrointestinal infections** are
- **viral,**
- **bacterial or**
- **parasitic**

Infections that cause gastroenteritis, an inflammation of the gastrointestinal tract involving both the stomach and the small intestine.

- Symptoms include:
- **Diarrhea,**
- **Nausea,**
- **Vomiting,**
- **Abdominal pain &**
- **Dehydration.**

- **Dehydration** is the main danger of gastrointestinal infections, so rehydration is important,
- **Most gastrointestinal infections are self-limited and resolve within a few days.**
- However, in specific populations **(newborns/infants, immuno-compromized patients or elderly populations), they are potentially serious. Rapid diagnosis**, **appropriate treatment** particularly important in these contexts.

Oral rehydration therapy (ORT)

- Replace water, salt, sugars lost due to diarrhea and vomiting
- The formula for the current WHO-ORS (known as ***low-osmolar ORS or reduced-osmolarity ORS***) per liter of fluid is:

Types of ORS??

- ❑ **Sodium bicarbonate based**
- ❑ **Trisodium citrate based**
- ❑ **Reduced osmolarity ORS**
- ❑ **Super ORS**

Sodium bicarbonate based ORS

Composition

Contents	(gm)
NaCl	3.5
Glucose	20.0
KCl	2.5
Sodium bicarbonate	2.5

ORS composition recommended by WHO and UNICEF

	Original ORS	Low osmolarity ORS*
Glucose, anhydrous	20	13.5
Sodium chloride	3.5	2.6
Trisodium citrate	2.9	2.9
Potassium chloride	1.5	1.5

Values are in grams to be dissolved in 1 litre of water before use.

*Low osmolarity ORS has been the recommended treatment since 2002, as it reduces the stool output, vomiting and the need for IVT compared to the original ORS formula.

WHO Drug Information Vol. 16, No. 2, (2002)

Enteric viral infections

- **Rotavirus**
- **Norovirus: infants**
- **Astrovirus: children , adults**
- **Enteric adenovirus: adults, children**

✓ The most common cause of gastrointestinal viral infections

✓ Characterized by moderate to severe vomiting followed by watery *diarrhea* and fever.

✓ Transmitted from water, vegetables contaminated by feces.

✓ No specific antiviral drug just conservative treatment Antipyretics for fever, antiemetics for vomiting and ORS for dehydration.

Bacterial gastrointestinal infections

Include foodborne infections and food poisoning.

- **Antibiotics not recommended unless infection causes bacteremia and septicemia or in debilitating patients**.
- Common sources of bacterial GI infections include:
- *Salmonella enterica (S. typhi, S paratyphi)*
- *E.coli (*enterotoxigenic, enteropathogenic)
- *Shigella* spp (Bacillary dysentery)
- *Campylobacter jejuni*
- *Vibrio cholera*
- *Clostridium difficile: antibiotic associated diarrhea*

- **Non-typhoid Salmonella spp.**

Ciprofloxacin 500 mg/12h for 5-7 days

Contraindicated in child below 14 years, breast feed & pregnant.

Ceftriaxone 1g IV/24h (adult dose). For 5 days

Safe child below 14 years, breast feed & pregnant.

- **E. coli**

(enterotoxigenic ETEC and enteropathogenic EPEC)

Traveler's Diarrhea

Ciprofloxacin 500 mg/12h for 1-3 days

Azithromycin 500 mg/24h for 1-3 days

Safe child below 14 years, breast feed & pregnant.

Rifaximin 200 mg/8h for 3 days

Shigella spp (WHO recommended)

Ciprofloxacin 500 mg/12h for 1-3 days

Second line treatmen

Ceftriaxone 1g IV/24h (adult dose

Azithromycin 500 mg/24h for 1-3 days

- **Campylobacter jejuni** (WHO recommended)

Azithromycin 500 mg/24h for 3 days

- **V. cholerae**

Doxycycline 300 mg orally single dose >8 years

Ciprofloxacin 500 mg/12h orally for 3 days.

Azithromycin 500 mg/24h orally for 3 days. (An appropriate alternative regimen for adults, including pregnant women, and children).

- **Pseudomembranous colitis** (antibiotic-associated diarrhea or ***C. difficile* colitis**), can occur following antibiotic treatment. When antibiotics are given, most of the resident intestinal bacteria are killed with sparing of some types like *Clostridia difficile*. The normally harmless *C. difficile* grow rapidly because of lack of competition with other flora, and produce toxins. These toxins damage the inner wall of the intestines and cause abdominal pain, fever and diarrhea. Severe cases can be life-threatening.

- **Treatment:**
  - ✓ Stop all antibiotics whenever possible.
  - ✓ Oral **metronidazole** 250 mg /6 hr (1st choice).
  - ✓ If metronidazole failed, give oral **vancomycin** solution 125 mg/6 hr

- ***Antibiotics likely cause diarrhea include:***

ampicillin, amoxicillin, cephalosporins, clindamycin.

Antibacterial for Treatment of Cholecystitis

- Meropenem, imipenem/cilastatin
- Piperacillin/ tazobactum.
- Tigecycline
- Cefazolin, cefepime ?
- Ciprofloxacin, levofloxacin

- Each plus metronidazole

- **Antibacterial for Treatment of Liver abscess & peritonitis**

- Piperacillin/ tazobactum
- Tigecycline
- Meropenem

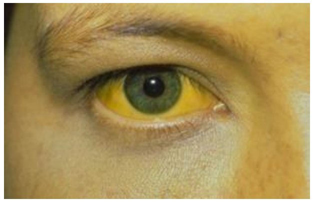
Treatment of Viral Hepatitis

Viral Hepatitis (Silent epidemic)

Viral hepatitis is caused by infection with any of :

- hepatitis A virus (HAV)
- HBV
- HCV
- hepatitis D virus (HD)
- hepatitis E virus (HEV)

Viral Hepatitis

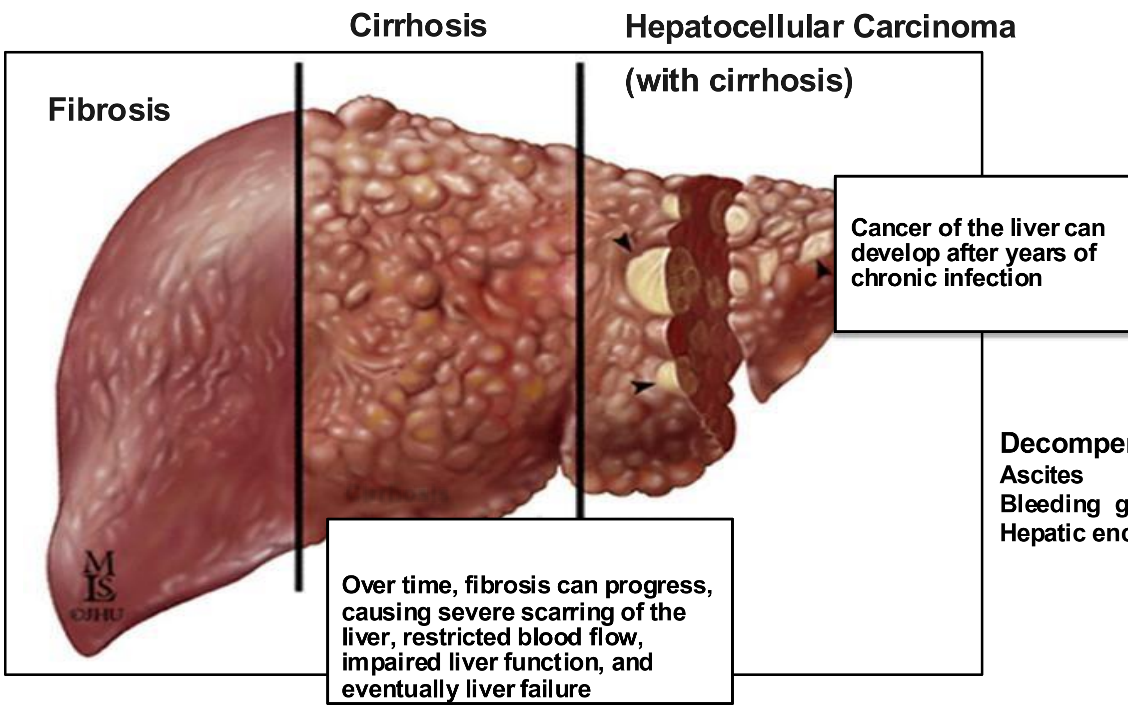


- All three of these unrelated viruses can cause severe illness in newly infected individuals, characterized by **nausea, malaise, abdominal pain, and jaundice.**
- Many new infections cause only mild symptoms or, in some cases, no symptoms at all.
- HBV and HCV can progress to chronic infections, but many who are chronically infected manifest no obvious signs or symptoms for decades— until they present with **cirrhosis, end-stage liver disease, or hepatocellular carcinoma.**

HBV, HCV

- Hepatitis B (DNA virus) and hepatitis C (RNA virus) are the most common causes of:
- Chronic hepatitis
- Cirrhosis
- Hepatocellular carcinoma

Chronic Viral hepatitis



Decompensated cirrhosis:
Ascites
Bleeding gastroesophageal varices
Hepatic encephalopathy Jaundice

Chronic hepatitis carriers remain infectious and may transmit the disease for many years

Diagnosis

Serology

HAV-Ab IgM

HBsAg HBeAg, HBcAb IgM

HCV Ab

HDV Ab IgM

HEV Ab IgM

PCR (Polymerase Chain Reaction)

Viral Hepatitis

Over View

Virus	Source	Transmission	Chronicity	Prevention
A	Stool	Feco-oral	No	Immunoglobulin Vaccination
B	Parentral	Percutanous per mucosal	Yes	Immunoglobulin Vaccination Immunoglobulin Vaccination
C	Parentral	Percutanous per mucosal	Yes	Blood donor screening Risk behavior modification
D	Parentral	Percutanous per mucosal	Yes	HBV Prevention
E	Stool	Feco-oral	No	Safe water cleaning

Treatment

Supportive management

Prevent infection spread

Specific antiviral therapy (may be considered for HBV & HCV)

Liver transplant for Fulminant hepatitis

Chronic HBV infection, treatment goals

The main goal of therapy for patients with chronic HBV infection is to:

- Improve survival and quality of life by preventing disease progression, and consequently HCC (hepatocellular carcinoma) development.
- ***Additional goals:***
- To prevent mother to child transmission
- Hepatitis B reactivation and prevention
- Treatment of HBV-associated extrahepatic manifestations.

Drugs for hepatic viral infections

Interferon-α
Pegylated interferon

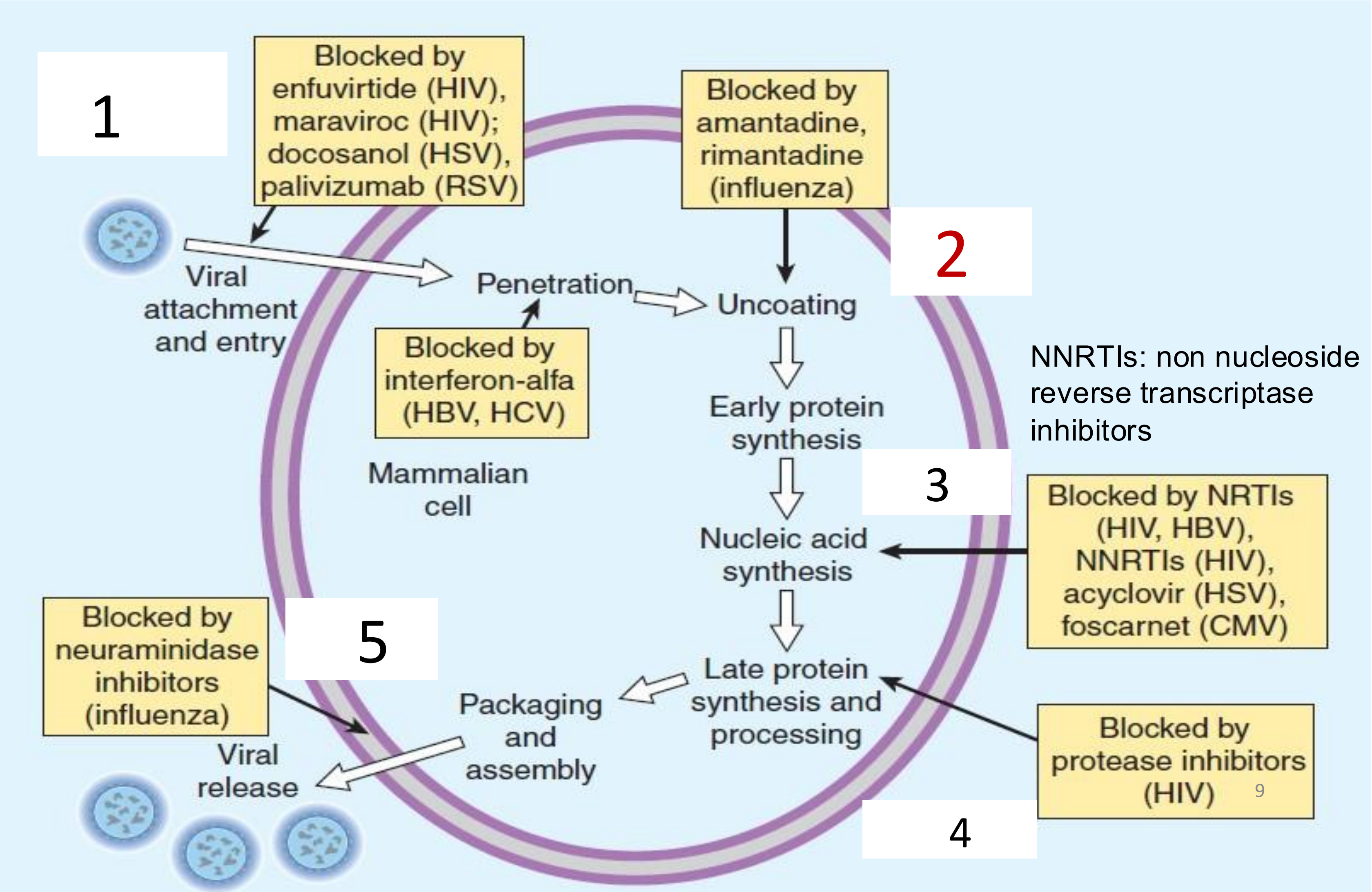
Nucleoside Analogues
Lamivudine
Entecavir
Telbivudine

Nucleotide analogues
Polymerase inhibitor
Sofosbuvir
Adefovir
Tenofovir

Protease Inhibitors
Boceprevir
Telaprevir
Atazanavir

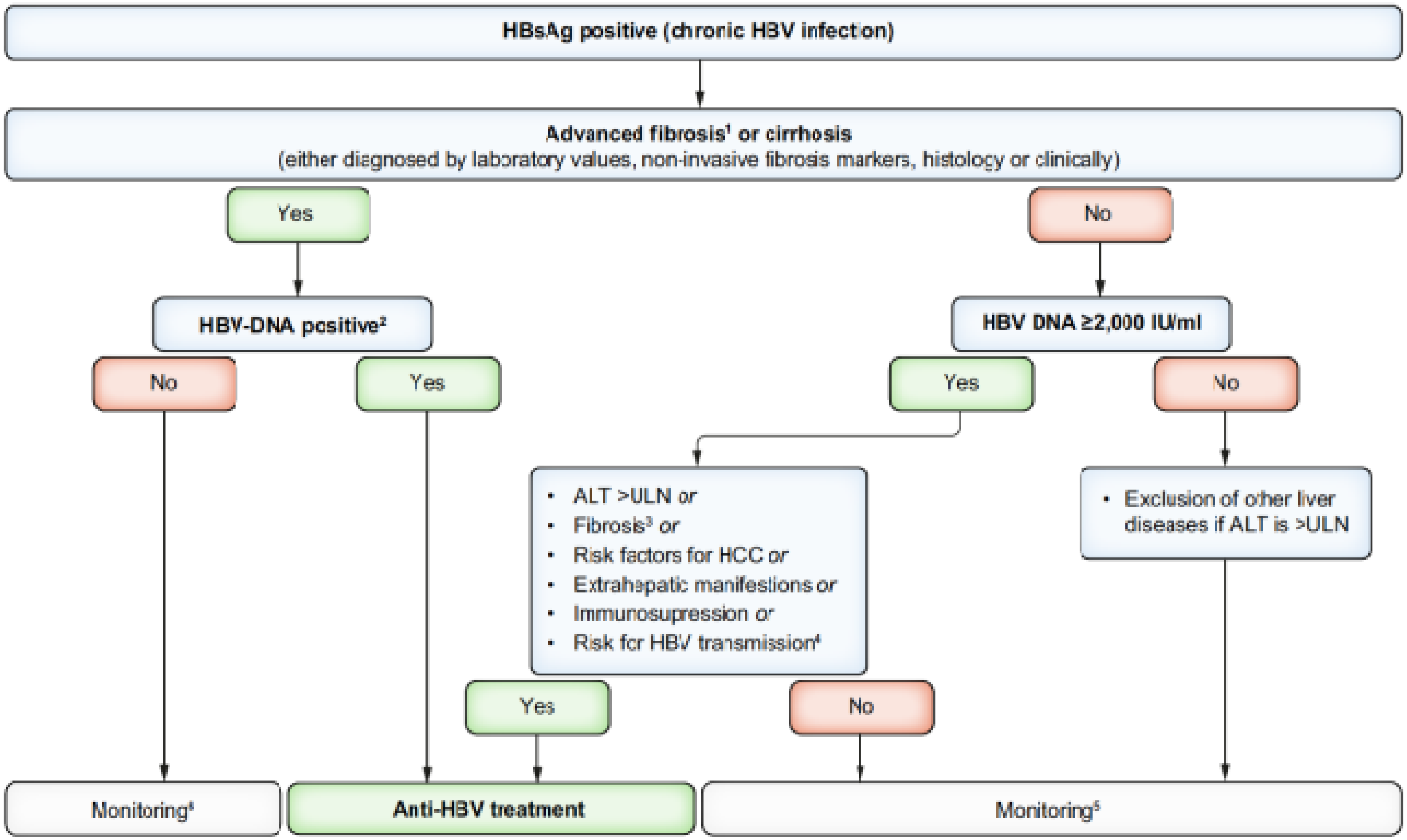
The goal of therapy for patients with HBV infection is to prevent the progression of liver disease to cirrhosis and hepatic cell cancer.

Site of action of anti viral agents used in the treatment of hepatitis virus



Steps in viral replication

Simplified Treatment Algorithm



European Association for the Study of the Liver (EASL) clinical practice guidelines on HBV

1) Interferon Alfa (IFN)

- **Interferon Alfa therapy – First Line**
- IFNs are a family of cytokines naturally produced by host cells in response to viral infections which trigger the protective defenses of the immune system
- Effective for both chronic HBV, HCV infections
- **Mechanism of action**:
- **Antiviral**: **Inhibition of viral penetration (mainly)**, uncoating, mRNA synthesis, and translation, and virion assembly and release
- Immunomodulatory
- Antitumour (hairy cell leukemi)

Pharmacology of IFN-alpha

- Is available as pINF-alpha (pegylated interferon-alpha: (IFN-α 2a or IFN-α 2b). SC, IM
- Pegylated: polyethylene glycol covalently attached to *interferon-α-2a* or *-α-2b:SC, IM*

pINF-alpha is Better Choice than IFN-Alpha:

- Greater Bioavailability:
- Increase the size of the molecule
- Delays absorption from the injection site
- Lengthens the duration of action of the drug
- Decreases its clearance
- Better treatment schedule (once wk vs three times per week for INF-α)

Pharmacology of pINF-alpha

- Bioavailability of both IFN- α (2a or 2b) is high (above 80%) when administrated IM or SC.
- Eliminated: principally via kidneys
- **Half-life:**
- IFN-α 2b: 40 h: more rapidly absorbed
- IFN-α 2a: 65 h; absorbed more slowly

INFs: Side effects

- "Flulike Symptoms",
- Transient live enzyme elevation
- Alopecia, rash, diarrhea" Subside with continued administration
- Depression: Caution in psychiatric, epilepsy, thyroid disease

Principal dose-limiting toxicities:

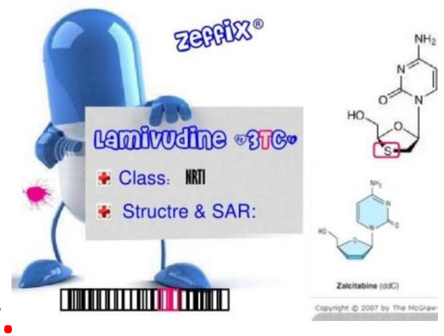
- Bone marrow suppression (BMS)
- Severe fatigue and weight loss
- Autoimmune disorders such as thyroiditis

Nucleoside Analogues:

- **Lamivudine, Entecavir, Telbivudine**
- **Mechanism of action**: inhibition of viral reverse transcriptase
- Phosphorylated by host cellular enzymes to the triphosphate (active) that prevents HBV DNA.
- By inhibiting viral reproduction, the immune system is able to clear the virus more effectively.
- Attain high intracellular conc. than in plasma

Lamivudine: Pharmacology

- Inhibitor of both HBV and HIV
- **MOA**: phosphorylated by host cellular enzymes to the triphosphate (active) form and competitively inhibits HBV RNA-dependent DNA polymerase and HIV reverse transcriptase
- **Given** PO once daily
- Rapidly absorbed; Bioavailability 82% in adults
- Diffuses freely across the placenta
- **Dose needs reduction in patients with renal insufficiency.**
- Hepatic impairment does not affect the pharmacokinetics
  - **Problem: High rates of resistant mutations**
  - **Side effect:** lactic acidosis; headache and dizziness



Ribavirin

- Guanosine nucleoside analog
- Inhibit viral RNA-dependent RNA polymerase of certain viruses, including influenza A and B, parainfluenza, respiratory syncytial virus, HCV, and HIV-1.
- **Pharmacology:**
- Oral bioavailability 64%:
- Increases with high-fat meals
- Decreases with co-administration of antacids.
- Given in combination with SC interferon alfa-2b for the treatment of chronic hepatitis C infection in patients with compensated liver disease

Ribavirin (Cont.)

Adverse effects

- Dose-dependent hemolytic anemia
- Depression, fatigue, rash, insomnia, nausea

- **Contraindications to ribavirin therapy:**
- Anemia
- end-stage renal failure
- severe heart disease
- pregnancy

Nucleotide analogues: adenosine analogue

MOA: inhibition of viral reverse transcriptase

- **Tenovir**
- Antiretroviral agent, licensed for patients with chronic HBV infection
    - Given: Once daily
    - Highly effective with low resistance than adenovir
- activity against lamivudine- and entecavir-resistant hepatitis virus isolates
- Side effects; nausea and bloating

Sofosbuvir

A nucleotide analog

Prodrug, triphosphorylated within the cells to produce its action.

The required enzymes for its activation are present in the human hepatic cells, converted to its active metabolite during the first-pass metabolism, directly at the desired site of action: The liver

SOF given as a part of a regimen including peg-IFN and ribavirin

excellent alterative in patients who are contraindicated to IFN therapy, or have stopped the IFN because of adverse effects

Sofosbuvir (Cont)

- Given orally as a single daily dose
- can be administered without the need of interferon therapy.
- No significant difference in the half-life with or without hepatic impairment
- No dosage adjustments are recommended for hepatic or mild-to-moderate renal impairment.
- **ADVERSE EFFECT:**
- small decrease in the Hb levels
- Headache, insomnia, fatigue, nausea, dizziness, pruritis
- Drug interaction:
- No clinically significant interactions between sofosbuvir wit immunosuppressive agents and Anti-HIV agnts.

Protease inhibitors

- **Telaprevir**

- Potent inhibitors of viral replication
- Oral antiviral agents for the adjunctive treatment of chronic HCV
- should be used in combination with *peg-interferon alpha* and *ribavirin* to improve response rates and reduce emergence of viral resistance
- Administered with food to improve absorption.
- Adverse-effects: anemia, rashes, lipodystrophy, liver toxicity, diabetes

- **Atazanavir**: has no effect on insulin sensitivity and serum lipid concentration. However, is associated with a significantly higher incidence of proximal tubulopathy.

Vosevi

- **Approved in July 18, 2017**
- Vosevi to treat adults with chronic hepatitis C virus (HCV) without cirrhosis (liver disease) or with mild cirrhosis.
- Vosevi is a fixed-dose, combination tablet containing:
- **Sofosbuvir, Velpatasvir, and *Voxilaprevir***
- **MOA:** block three proteins essential for the hepatitis C virus multiplication.
- **Sofosbuvir, Velpatasvir** : blocks the action of 2 different enzymes of viral RNA polymerase
- **Voxilaprevir**: block protease enzyme
- **Side effects**: headache, nausea, diarrhea
- **Interactions:** Vosevi must not be used together with: rosuvastatin, dabigatran, contraceptives, rifampicin, carbamazepine, phenobarbital, phenytoin.

References

- Lippincott Illustrated Reviews: Pharmacology; *7th ed* (2019) chapter 34 Antiviral Drugs by Elizabeth Sherman, pages 1251-1262.
- European Association for the Study of the Liver (EASL) clinical practice guidelines on HBV, 2025.
- EASL Recommendations on Treatment of Hepatitis C 2018q